Sohaib Ouakani

Computer Science & Engineering Graduate

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Education

Alma Mater Studiorum – Università di Bologna (Cesena Campus), Computer Science and Engineering (Ingegneria e Scienze Informatiche) — Cesena, Italy — Sept 2023 – July 2026

- Final grade: 110L
- Thesis: "Un linguaggio per domarli tutti: applicazioni full stack con librerie native in Kotlin Multiplatform"
- Advisor: Prof. Danilo Pianini · Co-advisors: Angela Cortecchia, Valter Venusti

Projects

Korsa — May 2026 – June 2026

University group project (3 people). Social running-tracker Android app, backed by a Supabase backend

- Built GPS run tracking as a foreground Android service (Fused Location Provider, GeoJSON routes rendered on an interactive MapLibre map)
- Implemented a social feed with likes, comments, following, and shareable deep links, using Supabase Realtime for instant updates
- MVVM architecture, with Koin DI, Jetpack Compose and Navigation Compose

FMI Workbench — Mar 2026 – June 2026

Full-stack Kotlin Multiplatform simulator for FMU (Functional Mock-up Unit) models

- First project in Kotlin, eventually became the thesis work
- Bound Kotlin/Native directly to the native FMILib C library via cinterop, with no glue code in another language across the stack
- Built a Ktor REST API (upload, info, simulate) consumed by a Compose Multiplatform frontend rendering results as line charts, with CI via GitHub Actions across Linux, macOS, and Windows

IoT Coursework — Dec 2025 – Jan 2026

University group project (4 people) for the Embedded Systems and IoT course: an Arduino game, an FSM-based drone hangar, and a 4-subsystem distributed system (MQTT, serial, HTTP) monitoring cistern water level and drainage control

- Built the game loop: sequence generation, input capture, difficulty potentiometer, LCD output, and win/lose state handling
- Implemented the drone hangar's core finite-state machine (flight/landing/hangar-door logic) on Arduino Uno, plus the sensor layer integrating ultrasonic, PIR, and LCD modules
- Built the Arduino-based drainage-control subsystem (state logic, messaging, logging) and the ESP32 hardware task for the tank-monitoring subsystem
- Developed part of the Python central control unit bridging serial and WebSocket communication, and the real-time JS dashboard rendering live water-level graphs

Skills

Programming Languages: Kotlin, Java, Python, C++, SQL, JavaScript, Rust

Mobile & UI: Jetpack Compose, Compose Multiplatform

Backend & Data: Ktor, REST APIs, Supabase (Auth, Postgrest, Realtime, Storage)

Systems & Tools: Kotlin/Native cinterop, CMake, Gradle, Docker, Git, GitHub Actions (CI/CD)

Testing: Kotest, Junit

Languages: Italian (Native), English (Fluent), Arabic (Native, Moroccan Arabic)